



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to SRI EZHUMALAIYAN AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

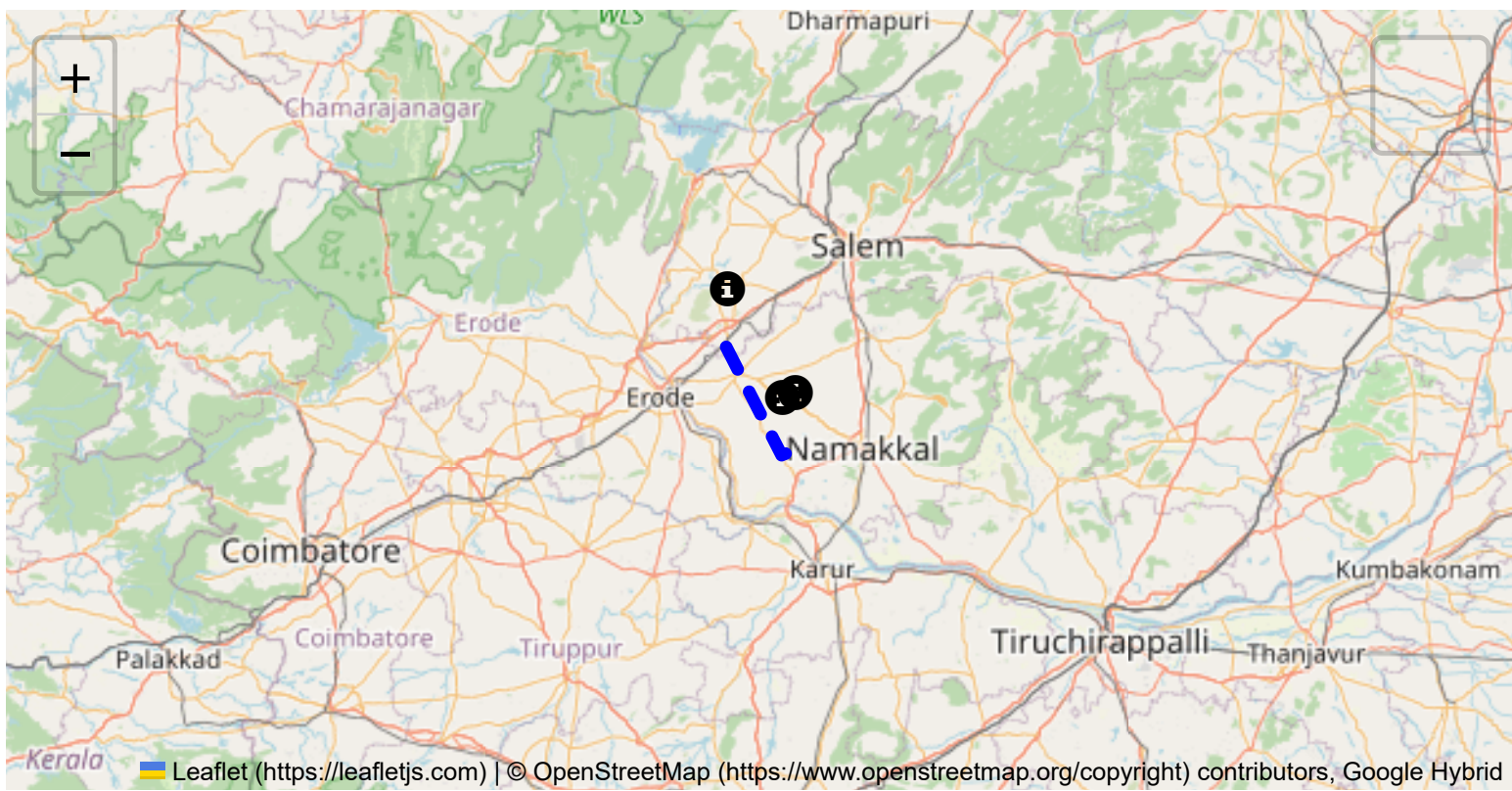
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

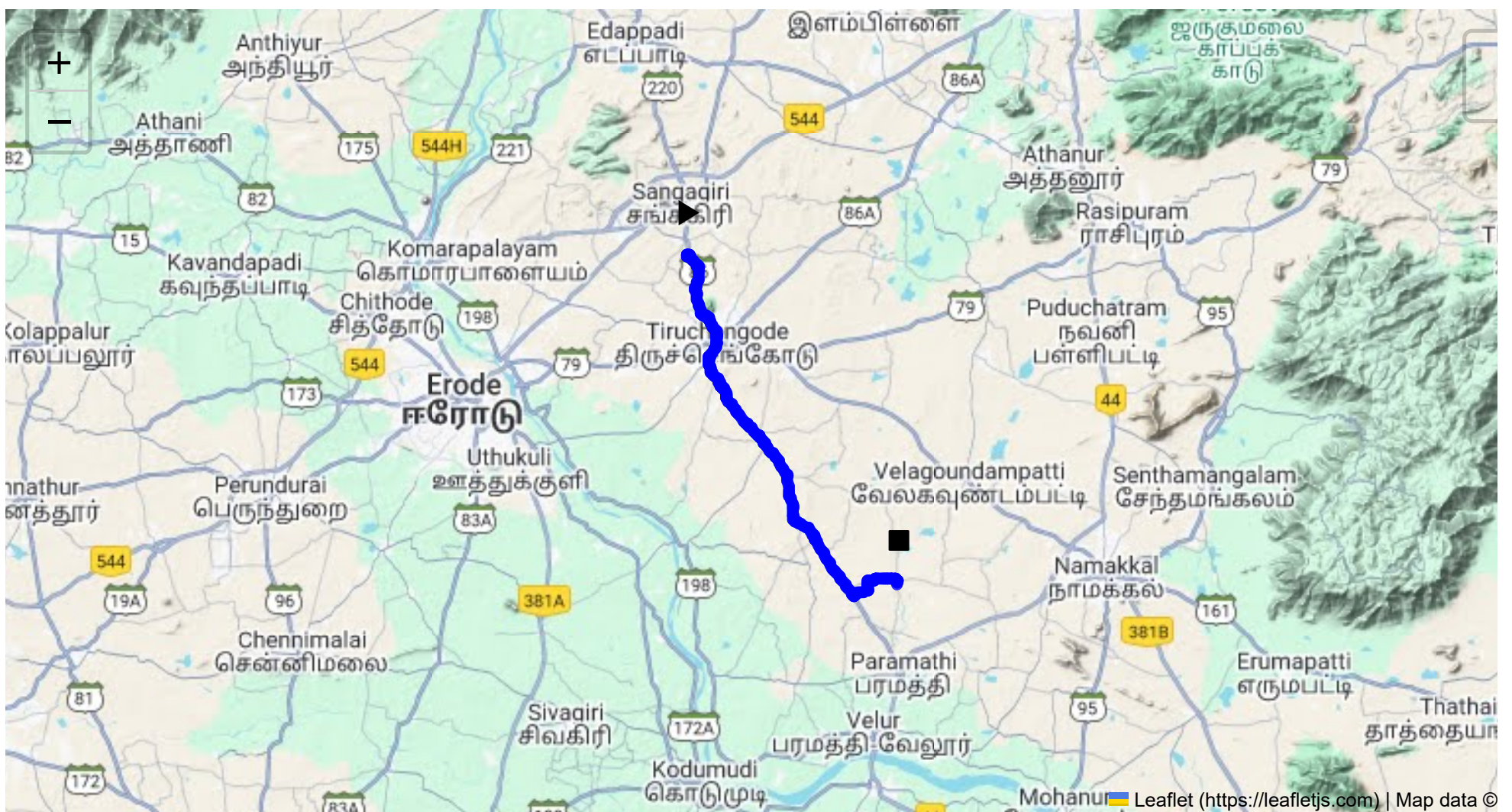
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 35.95 km
Estimated Duration: 0.8 hours
Adjusted Duration (Heavy Vehicle): 1.1 hours
Start: (11.4381, 77.8734)
End: (11.21389, 78.020332)



Welcome to the Journey Risk Management Study

To provide a detailed analysis of the route from Sangagiri to Paramatti via Vasanthapuram in Tamil Nadu, India, focusing on safety and risks for truck drivers transporting hazardous material, let's break down the various aspects outlined:

1. Overview of the Route Map:

- The route stretches approximately 36 kilometers, passing through rural and semi-urban areas, primarily using regional highways and roads. The main artery will likely include parts of the NH79 and SH86, which are common routes for transport in this region. The journey typically takes about 50 minutes for heavy vehicles under normal conditions.

2. Typical Weather Conditions and Potential Weather-Related Hazards:

- Tamil Nadu experiences a tropical climate with much rainfall during the monsoon season (June to September and Northeast monsoon from October to December). Heavy rains can lead to flooding and reduced visibility.
- During summer (March to June), the region can face high temperatures, potentially leading to road surface damage due to heat expansion.

3. Analysis of Traffic Patterns:

- Peak traffic hours generally occur in the morning (8:00-10:00 AM) and evening (5:00-8:00 PM) coinciding with commute times.
- Congestion is likely near town centers or where local roads intersect with the highway, such as entrances to Vasanthapuram and Paramatti.

4. Assessment of Road Quality and Infrastructure:

- The major highways in this region are reasonably well-maintained, but secondary and rural roads may have potholes or uneven surfaces.
- Watch out for narrow road segments and sharp turns, which may present difficulties for heavy vehicles, particularly in rural stretches.

5. Suggestions for Alternative Routes for Emergencies:

- In the event of road closures or emergencies, consider taking NH544 on the west side which runs parallel and could divert to Paramatti.
- SH86 is another potential bypass route if heavy congestion or accidents occur on the main route.

6. Summary of Local Regulations Affecting Hazardous Material Transport:

- Comply with Tamil Nadu's regulations on hazardous material transport, which require proper placards and adherence to specific transit schedules.
- Restrictions can include avoidance of certain areas during night hours and adherence to speed limits specially imposed for safety.

7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials:

- Historically, incidents mostly relate to vehicle breakdowns and overloading issues at weighbridge stations.
- Occasional spills or accidents highlight the importance of regular vehicle checks and the secure loading of hazardous materials.

8. Environmental Considerations and Sensitive Areas:

- Be conscious of agricultural areas, water bodies, and residential zones which may require slower speeds and more cautious driving to avoid contamination or disturbance.
- Consider the presence of wildlife crossings, especially in rural stretches.

9. Analysis of Communication Coverage:

- The route is moderately well-covered with cellular networks, but some rural segments might experience temporary dead zones.
- Ensuring a reliable GPS device with offline maps can help in areas with poor signal.

10. **Estimated Emergency Response Times for Different Route Segments:**

- Response times in urban and semi-urban areas like Sangagiri and Paramatti might be faster (approx. 15-30 minutes).
- In more isolated stretches, response times can extend to 45 minutes or more, depending on the nature of the emergency and weather conditions.

11. **Overall Summary of Risk Assessment:**

- The primary threats along this route are related to weather, traffic congestion during peak hours, and road surface conditions, particularly on less-traveled segments.
- By taking proactive measures such as monitoring weather forecasts, planning travel to avoid peak hours, ensuring vehicle readiness, and knowing alternative routes, risks can be significantly mitigated.
- Adherence to regulatory guidelines and maintaining open communication channels are crucial for safe transit of hazardous materials.

In conclusion, while the route poses typical risks found in rural transport settings, being well-prepared and informed can greatly enhance safety and efficiency in transporting hazardous materials across this region.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Turn	High	11.43811, 77.87348	15 KM/Hr
2	Turn	Medium	11.43956, 77.87340	30 KM/Hr
3	Turn	High	11.43968, 77.87345	15 KM/Hr
4	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
5	Turn	High	11.37868, 77.89498	15 KM/Hr
6	Turn	High	11.37850, 77.89453	15 KM/Hr
7	Blind Spot	Blind Spot	11.20589, 77.99001	10 KM/Hr
8	Turn	Medium	11.20907, 78.00001	30 KM/Hr
9	Turn	Medium	11.21344, 78.00061	30 KM/Hr
10	Turn	Medium	11.21352, 78.00057	30 KM/Hr
11	Turn	High	11.21371, 78.00021	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
12	Turn	High	11.21602, 78.00047	15 KM/Hr
13	Turn	Medium	11.21680, 78.00562	30 KM/Hr
14	Turn	High	11.21687, 78.00574	15 KM/Hr
15	Turn	Medium	11.21730, 78.00572	30 KM/Hr
16	Turn	High	11.21743, 78.00578	15 KM/Hr
17	Turn	High	11.21674, 78.02083	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	T.C.A Hospital Tiruchengode	11.3791885, 77.8965774	30 km/h	Medium
6	hospital	Soorya Multispecialty Hospital	11.3786429, 77.8931912	30 km/h	Medium
7	clinic	Kongu Nursing Home	11.3783065, 77.8961134	30 km/h	Medium
8	hospital	Tiruchengode Government Hospital	11.37645, 77.89426	30 km/h	Medium
9	hospital	Krishna Hospital, Namakkal	11.3754811, 77.8931817	30 km/h	Medium
10	hospital	Tirukumaran Hospitals	11.3782118, 77.8914933	30 km/h	Medium
13	police	கரட்டுப்பாளையம் காவல் நிலையம்	11.357586, 77.8922539	30 km/h	Medium
14	hospital	Goverment Hospital, Nallur	11.2679698, 77.946863	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
11	school	KSR Educational institution	11.3772013, 77.8908807	30 km/h	Medium
12	school	MDV School	11.3719843, 77.8915244	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.37868, 77.89498



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.37850, 77.89453



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.20589, 77.99001
